Which data modality recognizes a gene set? A domain-agnostic consilience engine

Software: consilience-engine

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Abstract

Background. Knowledge about any gene system is distributed across data modalities: sequence and domain architecture, expression across cell types, interaction, ontology, genetics, and disease. Integrative methods typically combine modalities to sharpen predictions; they rarely ask the prior question of *which* modality most specifically recognizes a given kind of biological entity, and no lightweight, domain-agnostic, permutation-tested tool for asking it is openly available.

Methods. We present *consilience-engine*, a keyless, reproducible, open-source Python tool that operationalizes a question rarely posed with a reusable instrument: which single data modality most specifically recognizes a given kind of biological entity. For any set of gene groups it builds a gene–gene similarity graph per modality and scores each group by *separation* (within- versus between-group similarity) against a permutation null. All four lenses (protein-domain structure, cell-type expression, molecular interaction, and biological-process ontology) use public, key-free endpoints.

Results. On a curated 410-gene, 49-component map of the nervous system, components partition into three classes with distinct diagnostic lenses: cell types (principally glia and vasculature) are recovered by expression, molecular families by structure, and functional or disease modules by interaction (each significant, permutation $z \geq 4$; neuronal cell types are the exception, spinal-motor routing to interaction and cerebellar significant but just short of the $z \geq 4$ pattern threshold on expression). Unsupervised clustering recovers the a-priori classes above chance without their labels (adjusted Rand index 0.25 nervous, 0.55 immune, both $p < 0.001$; the nervous value is a weak-to-moderate agreement). Run unchanged on a 132-gene immune map and a 64-gene cardiac map from configuration files alone, the tool reproduces the per-class mapping (92% specialized-lens accuracy in immune), indicating the pattern is not specific to one organ system. The new result is the operationalized lens-map principle and its label-free unsupervised recovery, not the individual biological facts, which serve only to validate the tool.

Conclusions. The tool is released open-source with deposited atlases and a time-sliced literature-based-discovery benchmark on which pre-2015 topology is rank-discriminative for post-2015 links (ROC-AUC 0.85–0.90). Practically, it indicates which assay most specifically resolves an uncharacterized gene set.

Keywords: data integration; consilience; multi-omics; gene sets; cell-type expression; protein domains; protein interaction networks; reproducibility; literature-based discovery

Introduction

A defining feature of modern biology is that what we know about any gene is *distributed*: its sequence and domain architecture live in structural databases, its expression in single-cell atlases, its partners in interaction databases, its roles in ontologies, and its clinical relevance in genetics and disease resources. The dominant response has been **integration**: combining modalities to sharpen a prediction (of function, disease association, or network membership). Integration implicitly assumes the modalities are partially redundant, more views of the same underlying biology.

We invert the question. Rather than asking how to *combine* modalities, we ask which modality is *most diagnostic* of a particular kind of biological entity, and whether the answer is systematic. If different classes of entity (cell types, molecular families, signaling modules) are legible to different data types, then the modalities are not redundant but **complementary in a structured way**, and the structure of that complementarity is itself a finding about how biological knowledge is organized. Answering this at scale requires a tool that treats any named gene set in any organ system uniformly, pulls each modality from public key-free sources so anyone can reproduce it, and tests every claim against a calibrated null rather than eyeballing a heatmap. No such lightweight, domain-agnostic, permutation-tested instrument is openly available.

This is distinct from established multi-omics integration, which fuses sample-level measurements to sharpen a downstream prediction: MOFA decomposes variance into shared and modality-specific factors (Argelaguet et al., 2018), similarity network fusion combines patient similarity networks (Wang et al., 2014), and mixOmics and DIABLO select correlated features across data blocks (Rohart et al., 2017; Singh et al., 2019). It is also distinct from gene-centric network-integration methods that fuse many gene networks to predict gene function or set membership, such as GeneMANIA (Warde-Farley et al., 2010) and the tissue-specific functional networks of GIANT (Greene et al., 2015), whose output is a ranked list of new members rather than the diagnostic modality itself. Those methods integrate modalities over samples or into one fused network; here the object is the gene-gene similarity graph per modality, and the question is which modality is diagnostic for which class of gene set, so the two lines are complementary rather than competing.

Here we present **consilience-engine**, an open-source Python tool that fills this gap. For any set of named gene groups it retrieves several public data modalities, turns each into a gene–gene similarity graph, and asks, per group and per modality, a single question: *are this group's genes more similar to each other than to everything else?* We call the answer the group's **separation** in that lens. Running the tool across a nervous-system gene set grown over four rounds to span the major transmitter systems, all glial and vascular cell types, electrical and synaptic machinery, sensory and developmental systems, and disease modules (410 genes in 49 components) produces a

clean and, we argue, general answer, which we then validate in two further organ systems (immune and cardiac).

The individual biological observations the tool recovers (that glia are defined by expression, that receptors share protein domains, that connectomes are small-world) are largely known. The contributions here are (i) a domain-agnostic, keyless, reproducible **software tool**; (ii) the **lens-map principle** (cell types→expression, families→structure, modules→interaction) and the structure–expression complementarity it reveals, with permutation-null-tested effect sizes; (iii) the demonstration that the principle **generalizes** across three organ systems; and (iv) the tool as a practical resource for experimental-design triage and forward discovery (Swanson, 1986). The individual biological facts the tool recovers are largely known and are used here only to validate it; the genuinely new results are the per-component diagnostic (which single lens most specifically recognizes each entity-class, tested against a calibrated null) and the label-free unsupervised recovery of the entity-classes (adjusted Rand index 0.25 nervous, 0.55 immune, permutation $p < 0.001$), which to our knowledge have not been reported for gene-set entity-classes in this form.

Materials and Methods

Gene sets and components. Human gene symbols were resolved to Entrez and Ensembl identifiers via MyGene.info (Wu, MacLeod & Su, 2013). The nervous-system atlas (410 genes, 49 components) was assembled over four rounds spanning transmitter systems, glia, vascular and ependymal cells, excitability, synaptic and plasticity machinery, sensory and developmental systems, and disease modules; the immune atlas (132 genes, 18 components) spans T, B, NK, myeloid, dendritic and neutrophil cell types, Toll-like/cytokine/chemokine/Fc receptor and MHC and complement families, and inflammasome/JAK-STAT/TCR-signaling/antigen-processing/IFN/NF-κB modules. Component membership is used only as ground-truth labels, never to build similarities.

Lenses. *Structure*: InterPro (Blum et al., 2025) domain accessions per gene from UniProt (reviewed, human), Jaccard similarity. *Expression*: Human Protein Atlas (Uhlén et al., 2015) per-gene cell-type RNA (field selected per tissue: brain single-nucleus nCPM for the nervous system, blood-cell nTPM for the immune system), log1p, cosine similarity. *Interaction*: STRING v12 (Szklarczyk et al., 2023) combined score (0–1) for all within-set pairs, one network call. *Process*: GO biological-process term identifiers per gene via MyGene, Jaccard.

Separation and null. For component C with gene set g in a lens with similarity s, separation = $\text{mean}_{i<j \in g}\, s(i,j) - \text{mean}_{i \in g,\, j \notin g}\, s(i,j)$, with missing pairs set to 0. The permutation null draws 2,000 random gene sets of size $|g|$ from the atlas; we report z = (observed − null mean)/null sd and empirical p. We verified that this permutation procedure is calibrated: on 400 random (true-null) gene sets spanning the component-size range (n = 6, 8, 12), the empirical permutation p-values are uniform (Kolmogorov–Smirnov vs. uniform, $p = 0.59$) and the false-positive rate is 0.045 at $\alpha = 0.05$, so the null controls Type-I error and does not manufacture significance. This calibration

establishes Type-I control only; it does not speak to power or to the curated-component non-independence noted in the Discussion. Because each component is tested across four lenses over 49 components, per-component best-lens significance is reported after Benjamini–Yekutieli FDR control (valid under arbitrary dependence, appropriate because the per-component separation statistics are coupled through a shared between-group similarity term and because the four lenses are mutually correlated); all 49 components survive at FDR < 0.05 (best-lens permutation $p \leq 0.008$). The calibration study uses 1,000 draws per null across its 400 true-null sets, and the unsupervised adjusted-Rand-index nulls use 5,000 label permutations. These budgets bound Monte-Carlo error: the calibration false-positive rate of 0.045 over 400 sets has a Monte-Carlo standard error of $\mathrm{sqrt}(0.045(1 - 0.045)/400) = 0.010$, about 0.5 standard errors from the nominal 0.05, and every per-component best-lens p (at most 0.008) and every adjusted-Rand-index p (below 0.001) lies more than two Monte-Carlo standard errors from its reporting threshold. The structure-versus-expression correlation ($r = -0.29$, $p = 0.043$) uses an analytic Pearson p, not resampling, so it carries no Monte-Carlo error; its sampling uncertainty is characterized by the bootstrap and design analysis in the Discussion. All permutation and label-permutation procedures use a fixed random-number-generator seed (1).

Unsupervised class recovery. Components were clustered by their z-scored four-lens separation profiles with class labels withheld; recovery of the a-priori classes was scored by the adjusted Rand index against a label-permutation null (5,000 permutations). Because the clustering algorithm is a nonequivalent analytic choice (Del Giudice & Gangestad, 2021), we report a method-invariance grid of four algorithms (k-means, Ward and average agglomerative, and a Gaussian mixture) at $k = 2, 3, 4$ rather than a single pick.

Consilience matrix. For each pair of modalities we compute the Spearman correlation of their gene-pair similarity values over shared pairs (threshold-independent); blocks are summarized as mean within- versus between-block correlation. The twenty gene-keyed modalities include literature co-occurrence, STRING (Szklarczyk et al., 2023), pharmacology (Harding et al., 2024), gene–disease and genetics (Ochoa et al., 2023), co-expression, KEGG and Reactome pathways, single-cell, brain-snRNA and bulk (GTEx Consortium, 2020) expression, proteomics, protein class, GO biological process, InterPro (Blum et al., 2025), co-essentiality, mouse-knockout and zebrafish phenotype (Bradford et al., 2022), and constraint.

Connectomes. Edge lists were obtained from openworm/ConnectomeToolbox (the *C. elegans* wiring diagram of White et al. (1986) as re-annotated by Cook et al. (2019)), the Janelia hemibrain public release (Scheffer et al., 2020), Netzschleuder (Peixoto, 2020; Ciona, *Drosophila* larva, macaque), the Allen Mouse Connectivity Atlas (Oh et al., 2014; rebuilt at 316 summary structures from the complete projection-unionize population), the Network Data Repository (Rossi & Ahmed, 2015; cat), and the Budapest reference connectome (human DTI). $\sigma = (C/C_{rand})/(L/L_{rand})$ (Humphries & Gurney, 2008), with C the mean clustering, L the mean shortest-path length (sampled for large graphs), and random-graph baselines from the Erdős–Rényi expectation;

modularity Q from greedy/Louvain (Blondel et al., 2008) community detection. Graphs were binarized; density matching keeps top-weight edges to a target density.

Literature-based discovery. PubMed MeSH co-occurrence graphs were built from papers through 2015; topological link-prediction (common-neighbour and Adamic–Adar (Liben-Nowell & Kleinberg, 2007); resource-allocation (Zhou, Lü & Zhang, 2009)) was scored against term pairs first co-occurring 2016–2023, with generic check-tag MeSH excluded.

Implementation. The tool is written in Python 3 (numpy, scipy, scikit-learn, networkx) and released as the `consilience-engine` package (Software availability). It is config-driven (one JSON per domain: components, expression field, species), path-independent, and caches all network pulls for offline, reproducible re-runs.

Results

A keyless multi-lens consilience engine

Figure 1 — The Consilience Engine: Each Entity-Class Is Diagnosed by the Lens That Separates It

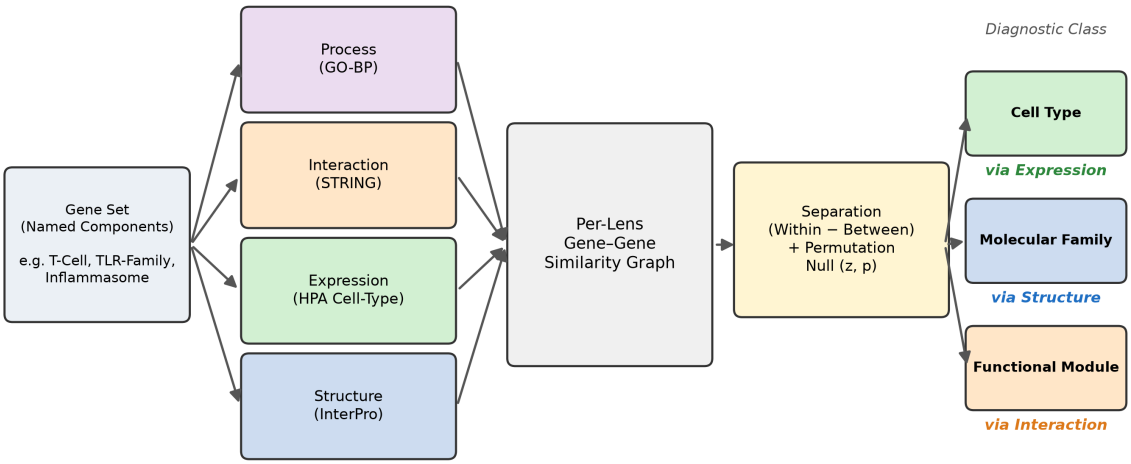


Figure 1 The consilience engine. A gene set is passed through four data lenses (structure, expression, interaction, process), each yielding a gene–gene similarity graph; per-component separation against a permutation null assigns each component its diagnostic lens.

The tool takes a dictionary of components (named gene groups) and, for each of four data lenses, builds a gene–gene similarity graph (Fig. 1): structure (Jaccard similarity of InterPro protein-domain annotations), expression (cosine similarity of cell-type expression vectors), interaction (STRING combined association score), and process (Jaccard similarity of Gene Ontology biological-process terms). For each component in each lens it computes separation and a permutation-null z-score and empirical p (Methods). All data sources are public and require no API key. The immune and cardiac demonstrations reproduce offline from the shipped caches with a single command (`scripts/repro.sh`); the nervous-system lens map and its permutation nulls are deposited as

precomputed results (`lens_map.json`, `neuro_perm_nulls.json`) produced by the same engine and lenses from the deposited 410-gene atlas and lens caches.

Each nervous-system entity-class has a distinct diagnostic lens

Figure 2 — Nervous System: Each Component Class Has a Distinct Diagnostic Lens

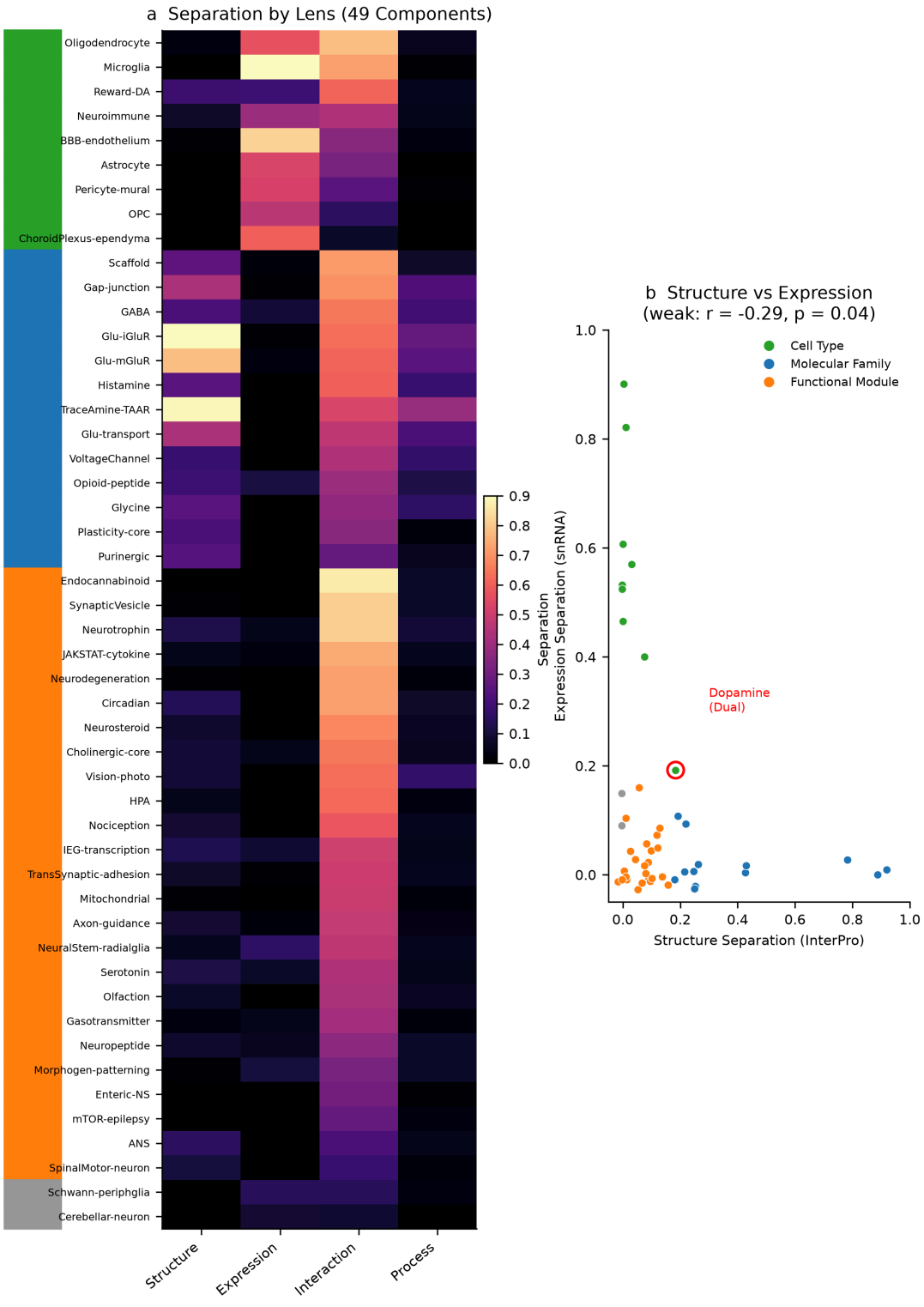


Figure 2 Nervous system (49 components). (a) Per-lens separation, components grouped by diagnostic class. (b) Structure- versus expression-separation: a weak

negative tendency ($r = -0.29$, $p = 0.043$) with a single dual case (dopamine).

Across the 49 nervous-system components, separation is strongly structured by lens (Fig. 2a). The interaction lens (STRING) has the highest separation for 38 of 49 components and detects nearly everything, so it is a near-universal detector rather than a partitioner; the diagnostic contrast is therefore between the two *specialized* lenses, structure and expression. Three patterns are individually highly significant (permutation $z \geq 4$, all $p < 0.01$):

- **Cell types separate on expression, not structure.** Microglia (snRNA separation 0.90, $z = 31$), blood–brain-barrier endothelium (0.82, $z = 32$), choroid-plexus/ependyma (0.61), astrocytes (0.53), pericytes (0.52), and oligodendrocyte precursors (0.47) separate sharply by cell-type expression and are essentially flat on structure (≈ 0). Glia are a *cell-identity* property, invisible to molecular-family lenses. Neuronal cell types are the exception: spinal-motor neurons do not separate by expression ($z = -0.3$) and route to interaction, and cerebellar neurons are significant on expression but fall short of the $z \geq 4$ pattern threshold ($z = 3.7$, $p = 0.008$), because neurons share a broad expression profile in the brain-expression lens.
- **Molecular families separate on structure, not expression.** Ionotropic glutamate receptors (InterPro 0.92), trace-amine receptors (0.89), metabotropic glutamate receptors (0.78), connexins (0.43), and the histamine/glycine/purinergic receptor groups ($\approx$0.25) separate by shared protein domain and are flat or negative on expression. They are all expressed in neurons, so the cell-type lens cannot distinguish them.
- **Functional and disease modules separate on neither specialized lens.** Synaptic-vesicle release (STRING 0.81), neurodegeneration (0.73), and the circadian clock (0.72) cohere through interaction while *failing* on both structure and expression. The failure itself is diagnostic that they are processes, not families or cells.

Per-component best-lens significance survives Benjamini–Yekutieli FDR control for all 49 components (best-lens permutation $p \leq 0.008$; Methods).

Structure and expression tend to be mutually exclusive. Plotting each component's structure-separation against its expression-separation (Fig. 2b) yields an L-shaped distribution: of 49 components, exactly one (dopamine/reward signaling) has both above 0.18 (it mixes a receptor *family*, DRD1–5 GPCRs, with dopaminergic-neuron *identity* markers, TH and DAT). We state this conservatively. The across-component anti-correlation is **weak**: Pearson $r = -0.29$ ($p = 0.043$, 95% CI [−0.53, −0.01]; robust to leave-one-out, with all 49 estimates negative at $r \in [-0.30, -0.27]$, and to correlation method, with all six Pearson/Spearman/Kendall specifications, with and without the dual-character component, negative and significant at median $r = -0.29$; Fig. 7). It is a real but modest negative *tendency* that explains <10% of variance, **not** statistical orthogonality, and is partly mechanical because many components are low on both lenses. It is computed from two purely data-driven lenses (InterPro domains and RNA expression), using neither the curated component labels nor text-mining.

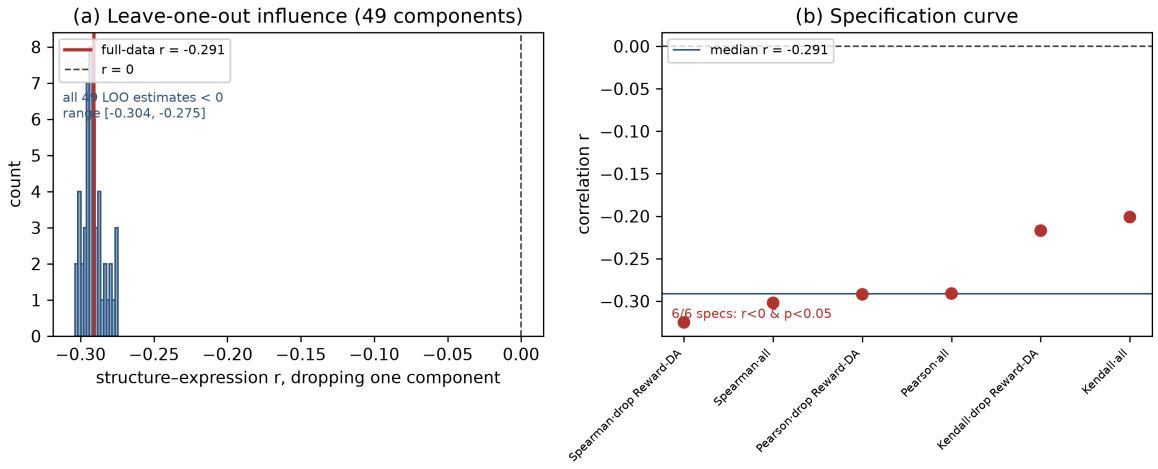


Figure 7 Robustness of the structure–expression relationship (nervous system, 49 components). (a) Leave-one-out influence: all 49 recomputed estimates remain negative (range [−0.30, −0.27]). (b) Specification curve: Pearson, Spearman, and Kendall correlations, with and without the dual-character component; all six are negative and significant, median r = −0.29.

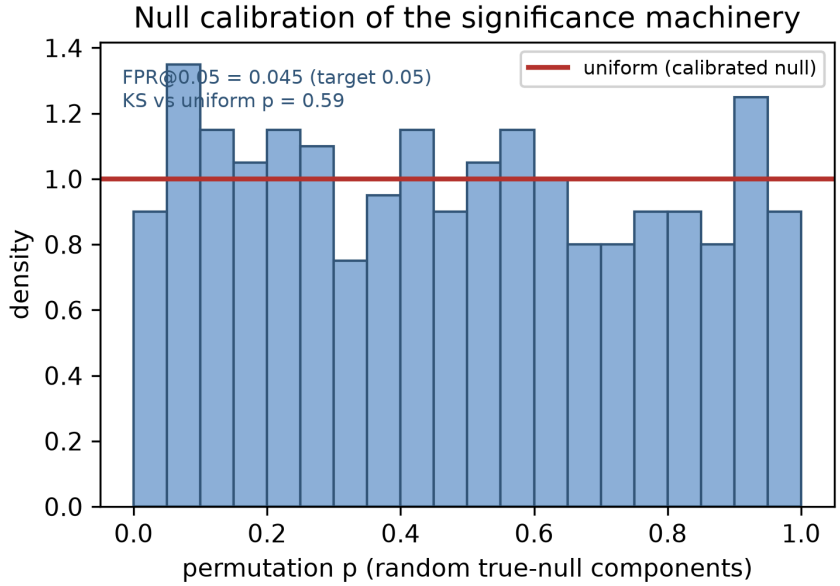


Figure 8 Calibration of the significance machinery. Permutation p-values for 400 random (true-null) gene sets are uniform (Kolmogorov–Smirnov vs. uniform p = 0.59), with a false-positive rate of 0.045 at α = 0.05.

The class structure is recoverable from the data, not just the labels. Because components are expert-curated, we tested whether the entity-classes emerge *without* the class labels: clustering components purely by their z-scored four-lens separation profiles (k = 3, labels withheld) recovers the a-priori classes above chance at adjusted Rand index **0.25** (nervous) and **0.55** (immune, below), both far exceeding the label-permuted null (permutation $p < 0.001$; $z = 8.4$ and 6.8); the nervous value of 0.25 is a weak-to-moderate absolute agreement, so significance here reflects reliable-above-chance recovery rather than strong recovery. A specialized-lens rule

(cell→expression, family→structure) classifies **64%** of nervous-system cell/family components correctly (the misses are dual-character or peripheral components the brain-expression lens cannot see). The recovery is not an artifact of one clustering choice: across a method-invariance grid of four algorithms at k = 2, 3, 4, the a-priori classes are recovered significantly against the label-permutation null in every cell whose k matches the three-class structure (nervous, 8 of 8 at k = 3 or 4, ARI 0.15–0.28; immune, 7 of 8, the single miss being average linkage at k = 3), with the published k = 3 value mid-range; we deposit the full multiverse (`method_invariance.json`) rather than a single pick (Del Giudice & Gangestad, 2021). The interaction lens (STRING) is the best lens for 38 of 49 components because it aggregates many channels including text-mining; it is a near-universal *detector* but a poor *partitioner* (community ARI ≈ 0.13). The discriminating, non-circular facts are that cell types go to expression and families to structure.

The per-component diagnostic reproduces in the immune system

Figure 3 — The Principle Generalizes: Entity-Classes Recover From Lens Data in a Second Organ System

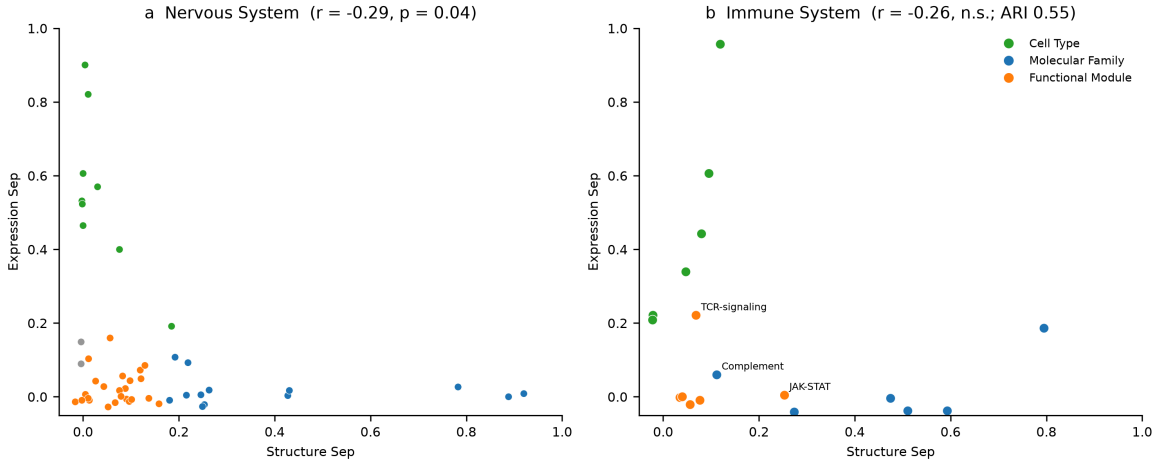


Figure 3 Generalization. Structure versus expression separation in the nervous (left) and immune (right) systems; cell types load expression, families load structure, modules sit near the origin. Immune classes recover unsupervised at adjusted Rand index 0.55.

To test whether the lens-map is a property of the nervous system or of biological data in general, we ran the identical tool, unchanged, on the immune system, swapping only the expression lens to HPA blood-cell RNA (132 genes, 18 components). The per-class mapping reproduced cleanly (Fig. 3): among the two specialized lenses, cell types separate on expression (B-cell 0.96, T-cell 0.61, NK 0.44) and families on structure (Toll-like receptors 0.79, z = 34; chemokine receptors 0.59; MHC 0.51; Fc receptors 0.47), while functional modules separate on neither and are detected only by interaction (inflammasome, antigen processing, NF-κB, IFN response). As in the nervous system, interaction has the highest separation for many components regardless of class, so the diagnostic contrast is between structure and expression. Unsupervised recovery reaches adjusted Rand index **0.55** (permutation $p < 0.001$, z = 6.8) and the specialized-lens rule classifies **11 of 12 (92%)** cell/family components correctly. The single exception, complement, is a structurally diverse

cascade that routes to interaction. Two further dual-character cases are equally principled illustrations, exactly analogous to dopamine in the brain: JAK-STAT is a module built from a kinase family (also structural), and TCR signaling is a module localized to T-cells (also expression). The across-component structure–expression anti-correlation shows the same *direction* as the nervous system ($r = -0.26$; leave-one-out $r \in [-0.35, -0.21]$) but, with only 18 components, does not reach significance ($p = 0.29$, 95% CI $[-0.65, +0.23]$); we report it as a consistent negative tendency, not a confirmed replication.

The central, defensible result is therefore the **per-component diagnostic and its unsupervised recovery across two organ systems** (cell types are read by expression, molecular families by structure, functional modules by interaction), rather than the weak across-component correlation. As a domain-agnosticism and reproducibility check, the same tool runs on a **third domain, cardiac (64 genes, 11 components), from a configuration file alone with no code changes**; the structure–expression tendency held again ($r = -0.28$) and, among the two specialized lenses, the clear cases separated as expected (adrenergic/cholinergic receptors and connexins on structure; endothelial and smooth-muscle cells on expression). The cardiac map is messier, because in the heart a cell's identity is largely its contractile module and "cardiac ion channels" span three unrelated structural families; this illustrates both the tool's reach and its limits.

Cross-modality agreement: three weak-to-modest axes

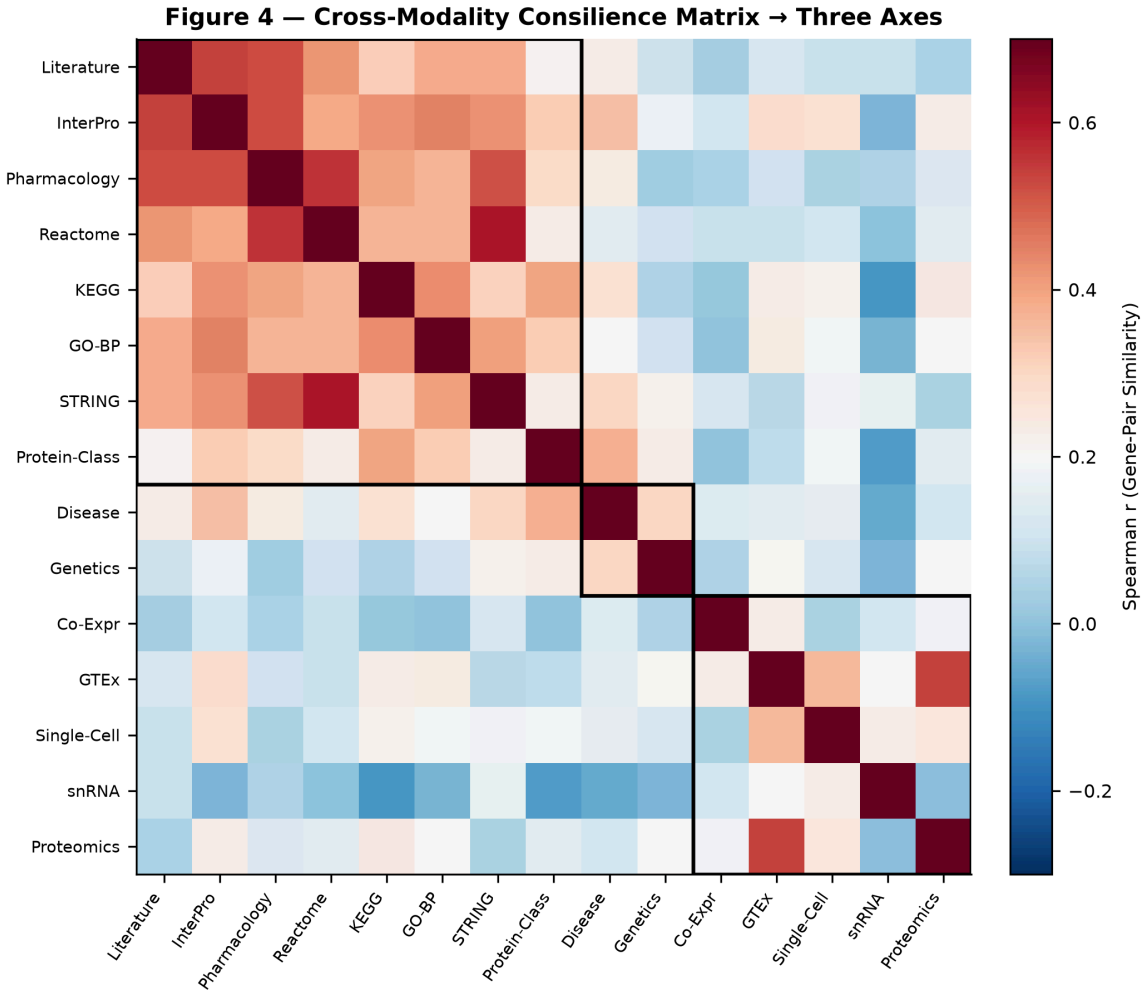


Figure 4 Cross-modality consilience matrix (Spearman correlation of gene-pair similarities) across 20 modalities, with the three axis-blocks outlined.

Beyond the four lenses used for the lens-map, we computed pairwise agreement among twenty gene-keyed modalities (Methods). Collapsing the matrix into blocks shows three organizing but unequal axes: a firm **function/structure** axis (within-block agreement 0.40), and weaker-but-distinct **disease/genetics** (0.30) and **expression** (0.18–0.22) axes, with between-block agreement 0.10–0.19 (Fig. 4). Three orthogonal data *types* added to probe for a fourth axis (co-essentiality, in-vivo phenotype, constraint) formed none. This modality-level result mirrors the component-level lens-map: the same separation between *what a molecule is*, *where it is expressed*, and *what it does* recurs whether one interrogates data sources or biological entities.

The tool extends to connectomes: conserved modular small-world organization

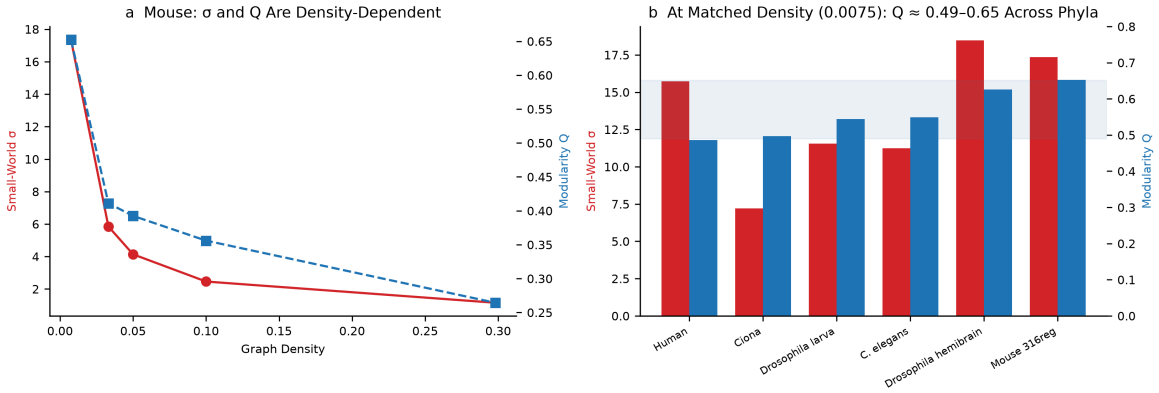


Figure 5 Connectome organization. (a) σ and Q are density-dependent (mouse density sweep). (b) At matched density, modularity Q ≈ 0.49–0.65 from worm to human (a supporting reproduction, not a central claim).

Because the tool is graph-based, the same topological analysis applies to wiring diagrams. We assembled connectomes from seven datasets (six species; two *Drosophila* developmental stages) across three phyla from public sources and computed the small-world coefficient σ and modularity Q. Because σ and Q are density-dependent (Fig. 5a), a fair comparison requires matching density. At a matched density of 0.0075, every connectome in the core set is strongly small-world (σ ≈ 7–18) and modular, with Q in a tight band of **0.49–0.65 from nematode to human** (Fig. 5b). One independent-lab macaque graph is excluded because it disconnects into an uninterpretable graph at this density. Matched-density thinning is aggressive: several retained graphs (the tunicate Ciona keeps 62% of nodes in its giant component, mouse and *C. elegans* ~83%) are sparse (mean degree ≈ 1.5–2), so the reported σ and Q band is approximate and its lower end (Ciona) is a partly fragmented graph. This is a *reproduction* of a known property (connectome small-worldness; Watts & Strogatz, 1998; Humphries & Gurney, 2008) using a keyless, matched-density pipeline with stated caveats (node count and edge semantics are not matched; matched-density thinning fragments the sparser graphs; there is no null model beyond σ's normalization), not a new discovery. It shows that the tool operates identically on a third class of nervous-system data.

Forward use: time-sliced literature-based discovery

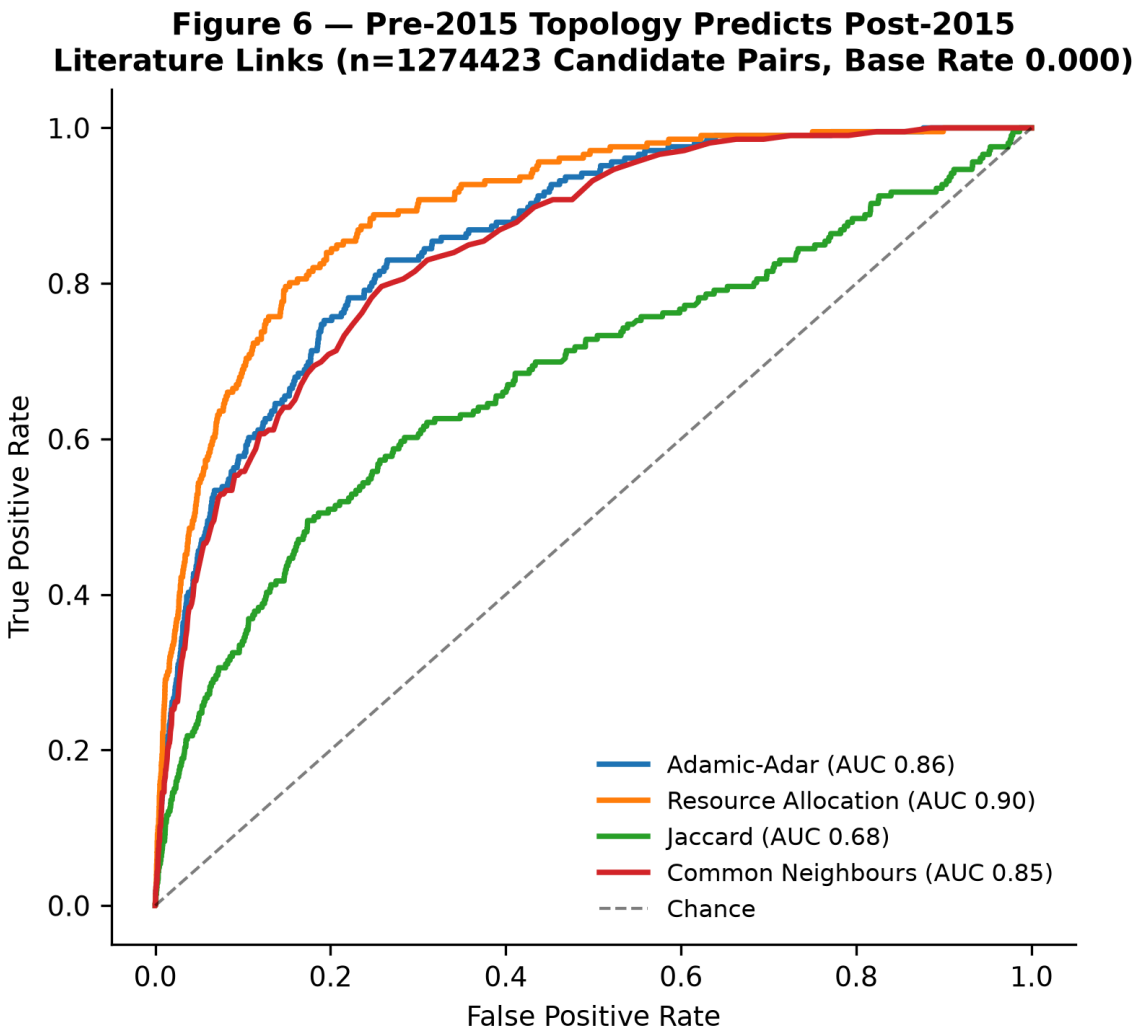


Figure 6 Forward use. Pre-2015 network topology ranks post-2015 literature co-occurrences above chance (ROC; resource-allocation AUC 0.90) over ~1.3M candidate term pairs; top-of-list precision is low (see text).

To show the tool supports hypothesis generation rather than only description, we implemented a time-sliced literature-based-discovery benchmark on real dated PubMed data: build a MeSH co-occurrence graph from papers published through 2015, and test whether its topology predicts term pairs that first co-occur in 2016–2023 (Swanson, 1986). Over ~1.3M candidate term pairs, the top link predictors (common-neighbour, Adamic–Adar (Liben-Nowell & Kleinberg, 2007), resource-allocation (Zhou, Lü & Zhang, 2009)) achieve ROC-AUC 0.85–0.90 (resource-allocation 0.90; Jaccard lower at 0.68). We report the precision side honestly, because the task is extremely imbalanced (230 positives among ~1.3M candidates, base rate ≈ 0.0002): ROC-AUC measures rank-discrimination, not usable top-of-list precision. The best predictor (resource-allocation) reaches only precision@100 ≈ 1% and PR-AUC ≈ 14× the base rate (`lbd_precision.json`); the others are near-chance at the top of the ranking. Pre-2015 topology therefore carries real but weak predictive signal about future co-occurrence: it demonstrates that the same graph primitives extend to forward prediction, but at these precisions it is a hypothesis-narrowing aid, not a high-yield discovery engine.

Discussion

The recurring result is that biological knowledge is *partitioned by entity-class across data modalities*. Cell types are encoded in expression, molecular families in structure, and functional modules in interaction. The two specialized encodings (structure and expression) tend to be mutually exclusive in practice, so a component is typically one kind of thing or another. This is why the modality that most cleanly partitions a set is class-dependent, and why integration adds value not as a noise-reduction step but because the modalities carry *categorically different* information. The pattern is not specific to the brain: the per-component diagnostic and its unsupervised recovery hold in the immune system, though we are careful that the *across-component correlation* is weak and, in immune, not individually significant; the cross-system claim rests on the per-component mapping and unsupervised class recovery, not on the correlation.

The practical value follows directly. **Experimental-design triage**: given an uncharacterized gene set, the tool indicates which assay will resolve it (expression separation → cell type, do single-cell work; structure → molecular family, do domain or structural analysis; only interaction → functional module, do co-IP or network experiments). **Target-class characterization**: applied to drug-target groups, the tool classifies a target's evidence type, relevant to repurposing and class-effect reasoning. **Forward discovery**: the link-prediction ranks candidate links above chance, so it can narrow, though at low top-of-list precision not confirm, hypotheses for any field.

Limitations. First, and most important, the across-component structure–expression anti-correlation is weak and underpowered: $r = -0.29$ ($p = 0.043$) in the nervous system and $r = -0.26$ (not significant, $p = 0.29$) in the immune system, and the mutual-exclusivity contingency test is not significant in either (Fisher $p = 0.41$, 0.15). The *direction* is nonetheless robust: every leave-one-out estimate remains negative and all six correlation specifications are negative and significant (Fig. 7). A design analysis makes the underpowering concrete: at the observed effect (assumed true $r = -0.29$, power 0.53) the sign is reliable and inflation modest (Type-S < 0.001, Type-M $\approx 1.3\times$), but if the true effect were half that ($r = -0.1$) the same $n = 49$ gives 10% power, a 4% sign-error rate, and a $\approx 3.4\times$ exaggeration ratio, so the reported magnitude should be read as a likely over-estimate; a 10,000-resample percentile bootstrap 95% CI is $[-0.41, -0.20]$ and the analytic Fisher-z interval is $[-0.53, -0.01]$; the two agree in sign, but the percentile bootstrap is markedly narrower (it is anticonservative for a correlation at $n = 49$), so we anchor inference on the wider analytic interval, which reaches -0.01, and on the marginal $p = 0.043$. We therefore do **not** claim statistical orthogonality; the defensible claims are the per-component diagnostic ($z \geq 4$, all $p < 0.01$) and the unsupervised recovery of the entity-classes (ARI 0.25 nervous, 0.55 immune). Second, component definitions are expert-curated, and 'molecular family' is operationalized as a group sharing protein-domain architecture, so some curated groups (for example neurotransmitter systems) are functional rather than strict sequence families and the specialized-lens accuracy is reported under this operational labeling; several lenses (literature, pharmacology, partly STRING via text-mining) draw on overlapping curated knowledge; we mitigate this by using components only as labels (never to build similarities; verified by an independent 18-agent adversarial audit), by computing the

structure–expression relationship from two purely data-driven lenses, and by the label-free unsupervised control. Third, the individual biological findings are confirmatory; the novelty is the tool, the operationalized lens-map, and the cross-system generalization, with forward LBD as the path to new biology; the connectome and consilience-matrix sections are supporting demonstrations of generality, not central claims. Fourth, the metrics have characteristic caveats: separation z rises with group size; the brain-expression lens is blind to peripheral systems (retina, olfactory epithelium, PNS); a group that secretly contains several cell types fails to separate; and connectome comparisons match density but not node count or edge semantics. An adversarial audit during development caught and fixed a data-truncation bug (an Allen connectivity pull silently capped at 500,000 of 756,833 rows), which we note as evidence of the verification discipline applied throughout.

Conclusions

`consilience-engine` operationalizes a simple question, which data modality recognizes a given kind of gene set, into a keyless, reproducible, permutation-tested tool. Across three organ systems it recovers a stable lens-map (cell types by expression, molecular families by structure, functional modules by interaction) that is present in the data independent of the curated labels, and, run forward on dated literature, its topology ranks future links above chance (though at low top-of-list precision). The tool is open-source and fully reproducible, and is intended as a practical instrument for experimental-design triage and forward discovery.

Software availability

- **Project name:** consilience-engine
- **Project home page:** https://github.com/patrickiben/consilience-engine (archived snapshot including code and data: OSF, DOI 10.17605/OSF.IO/ZKTRJ)
- **Operating system(s):** Platform independent
- **Programming language:** Python ≥ 3.11
- **Other requirements:** numpy, scipy, scikit-learn, networkx
- **License:** MIT (Open Source Initiative approved)
- **Version:** v1.0.0 (git tag `v1.0.0`), maintained under git version control at the project home page and archived as OSF DOI 10.17605/OSF.IO/ZKTRJ
- **Restrictions to use by non-academics:** None; the software is free to all users and requires no personally identifiable information.

The package ships example configurations for three systems (nervous, immune, cardiac); the immune and cardiac demonstrations reproduce offline from the shipped caches via a one-command reproduction script (`scripts/repro.sh`), and it ships an offline unit test and metamorphic and property-based tests (dilution monotonicity, lens-duplication and relabeling invariance) run under continuous integration. All four lenses use public, keyless endpoints. The connectome (small-

world/modularity) and literature-based-discovery analyses use the same graph primitives but are one-off analysis scripts rather than packaged engine features; their derived inputs and outputs are deposited (Data availability).

Data availability

The two curated atlases (nervous-system, 410 genes/49 components; immune, 132 genes/18 components), the cardiac demonstration set, all per-lens gene–gene similarity matrices, the lens maps, the consilience matrix, the connectome metrics, and the literature-based-discovery outputs, together with the source code, are archived on the Open Science Framework (OSF), DOI 10.17605/OSF.IO/ZKTRJ, under a Creative Commons CC-BY 4.0 license. All primary data are public and require no API key, and were accessed in 2026 from PubMed/E-utilities, STRING (v12), the Human Protein Atlas, InterPro/UniProt, MyGene.info, IUPHAR/BPS Guide to Pharmacology, Open Targets, KEGG, Reactome, GTEx, openworm/ConnectomeToolbox, the Janelia hemibrain release, Netzschleuder, the Allen Mouse Connectivity Atlas, the Network Data Repository, and ZFIN.

Declarations

Competing interests. The author declares no competing interests.

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Author contributions. Patrick Iben is the sole author: conceived and designed the study, wrote the software, performed the analyses, prepared the figures, and wrote and reviewed the manuscript.

Ethics. Not applicable; the study uses only public, aggregate, non-human-subject data.

Disclosure of use of AI-assisted tools including generative AI. A generative-AI coding assistant (Claude Code, Anthropic; model Claude Opus 4.8) was used by the author, Patrick Iben, between June and July 2026 to assist in drafting and editing the manuscript text, implementing the analysis pipeline, generating the figures, and verifying reference metadata. The author conceived the study, its analyses, and its interpretations, independently re-derived all numerical results, applied an adversarial code audit before submission, reviewed and validated every output, and takes full responsibility for the content. The AI-assisted analysis and figure-generation code is archived in the deposit (OSF, DOI 10.17605/OSF.IO/ZKTRJ); no AI tool was used to create or modify figures in any way that adds, removes, or alters the relationship of elements within an image; the figures were produced programmatically by the deposited figure-generation scripts (`code/figures/` for F1–F6, `results/` for F7–F8) from the deposited results. AI tools do not meet the criteria for authorship and are not listed as authors.

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